

(b) Let $x = 0$ and $y = \sqrt{2}$. Then, x is rational and y is irrational.

$\therefore xy = 0 \times \sqrt{2} = 0$, which is rational.

Hence, xy is not irrational.

(c) As shown in (b), xy is not necessarily irrational.

(d) $x + y$ is necessary irrational. But xy can be either rational or irrational.

Hence, (d) is true.

41. Let x and $(x + 1)$ be two consecutive integers. Then
 $(x + 1)^2 - x^2 = (x + 1 + x)(x + 1 - x) = (x + 1 + x) \times 1$
 $= (x + 1 + x) = \text{sum of given numbers.}$

42. If a and b are two rational numbers, then $\frac{a+b}{2}$ is a rational number lying between a and b .

43. $B > A \Rightarrow A < B \Rightarrow (A - B) < 0$.

Since A and B are both positive integers, we have
 $(A + B) > 0$ and $AB > 0$.

If $(A = 1 \text{ and } B = 2)$, we have $AB < (A + B)$.

If $(A = 2 \text{ and } B = 3)$, we have $(A + B) < AB$.

Thus, we cannot say which one of $A + B$ and AB has the highest value.

44. $0 < x < 1 \Rightarrow x^2 < x < 1 \quad \dots(i)$
 $\Rightarrow \frac{1}{x^2} > \frac{1}{x} > 1 > x > x^2 \quad [\text{using } (i)]$

Hence, $\frac{1}{x^2}$ is the greatest.

45. $p < 1 \Rightarrow \frac{1}{p} > 1 \Rightarrow \frac{2}{p} > 2 \Rightarrow \frac{2}{p} - p > 2 - p > 0 \quad [\because p < 1]$

Hence, $\left(\frac{2}{p} - p\right)$ is a positive number.

46. $(x^2 + x + 1) = \left(x^2 + x + \frac{1}{4}\right) + \frac{3}{4}$
 $= \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4} \quad \left[\because \left(x + \frac{1}{2}\right)^2 \geq 0\right]$

Hence, $(x^2 + x + 1)$ is greater than or equal to $\frac{3}{4}$.

47. $\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{7}\right) + \frac{1}{n} = \frac{(21 + 14 + 6)}{42} + \frac{1}{n} = \left(\frac{41}{42} + \frac{1}{n}\right)$.

This sum is a natural number when $n = 42$.

So, each one of the statements that 2 divides n ; 3 divides n and 7 divides n is true.

Hence, $n > 84$ is false.

48. $\frac{(16n^2 + 7n + 6)}{n} = \left(\frac{16n^2}{n} + \frac{7n}{n} + \frac{6}{n}\right) = \left(16n + 7 + \frac{6}{n}\right)$.

For $\left(16n + 7 + \frac{6}{n}\right)$ to be an integer, we may have $n = 1$

or $n = 2$ or $n = 3$ or $n = 6$.

Hence, 4 values of n will give the desired result.

49. $(p > q \text{ and } r < 0) \Rightarrow pr < qr$ is true.

50. $(X < Z \text{ and } X < Y) \Rightarrow X^2 < YZ$.

51. $x + y > p$ and $p > z \Rightarrow x + y > z$.

52. $\frac{(a-b)}{3.5} = \frac{4}{7} \Rightarrow (a-b) = \frac{4}{7} \times \frac{7}{2} = 2 \Rightarrow b < a$.

53. $\frac{13}{1} = \frac{13w}{(1-w)} \Rightarrow \frac{w}{(1-w)} = 1 \Rightarrow w = 1 - w \Rightarrow 2w = 1 \Rightarrow w = \frac{1}{2}$.
 $\therefore (2w)^2 = 4w^2 = 4 \times \frac{1}{4} = 1$.

Questions 54 to 57

Let the digits of the number in order be A, B, C, D, E .

Then, $A > 5 \Rightarrow E = 1$.

A, B, C are all odd and none of the digits is 3 $\Rightarrow A, B, C$ are the digits from 5, 7, 9.

Since A is the largest digit, so $A = 9$.

$\therefore B = 5$ or 7 and $C = 5$ or 7 .

Now, the number DE is the product of two prime numbers.
 $E = 1$ and $D = 2, 4, 6$ or 8 .

41 and 61 are prime numbers and 81 cannot be expressed as product of two primes.

Only 21 can be expressed as the product of two prime numbers ($21 = 3 \times 7$).

So, $D = 2$.

Hence, the number is 95721 or 97521.

54. The second digit of the number is either 5 or 7.

55. The last digit of the number is 1.

56. The largest digit in the number is 9.

57. The number is odd. So, it is not divisible by 2 or 4.
Sum of digits = $9 + 7 + 5 + 2 + 1 = 24$, which is divisible by 3 but not by 9.

So, the given number is divisible by 3.

58. The least prime number is 2.

59. **Statement 1.** Let $x = 4$ and $y = 15$. Then, each one of x and y is a composite number.

But, $x + y = 19$, which is not composite.

$\therefore$ Statement 1 is not true.

Statement 2. We know that 1 is neither prime nor composite.

$\therefore$ Statement 2 is not true.

Thus, neither 1 nor 2 is correct.

60. Prime numbers between 0 and 50 are:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43 and 47.

Their number is 15.

61. Each of the required numbers must divide $(143 - 3) = 140$ exactly.

Now, $140 = 5 \times 7 \times 4$.

Hence, the required prime numbers are 5 and 7.

62. Sum of first four prime numbers = $(2 + 3 + 5 + 7) = 17$.

63. Sum of all prime numbers from 1 to 20 = $(2 + 3 + 5 + 7 + 11 + 13 + 17 + 19) = 77$.

64. Clearly, 11 is a prime number which remains unchanged when its digits are interchanged. And, $(11)^2 = 121$.

Hence, the square of such a number is 121.

65. Let the required prime number be p . Let p when divided by 6 give n as quotient and r as remainder. Then
 $p = 6n + r$, where $0 \leq r < 6$

Now, $r = 0, r = 2, r = 3$ and $r = 4$ do not give p as prime.

$\therefore r \neq 0, r \neq 2, r \neq 3$ and $r \neq 4$.

Hence, $r = 1$ or $r = 5$.

66. Clearly, $21 = 3 \times 7$, so 21 is not prime.

67. Clearly, 19 is a prime number.

68. We know that 115 is divisible by 5. So, it is not prime.

119 is divisible by 7. So, it is not prime.

$127 < (12)^2$ and prime numbers less than 12 are 2, 3, 5, 7, 11.

Clearly, 127 is not exactly divisible by any of them. Hence, 127 is a prime number.